Master's Comprehensive Examination

Part I Theory (90 minutes) March 6, 2004

Instructions: Answer all four questions. Open book and open notes.

1. Let $X \sim B(2, 1/2)$. Let $Y \sim B(3, 1/3)$. Assume X and Y are independent.

(a) Find $P\{X \geq 1\}$.

(b) Find $P\{X \geq 2 | X \geq 1\}$.

(c) Find $P\{X = 2, Y = 3\}$.

(d) Find $P\{X = 2 | Y = 3\}$.

(e) Make a two way table whose entries are the joint probability distribution of X, Y, i.e., $P\{X = x, Y = y\}$, $x = 0, 1, 2$; $y = 0, 1, 2, 3$.

(f) Find $P\{X = 1 | X + Y = 2\}$.

2. Let X_1 and X_2 be normal variables with means $\mu_{x_1} = 1$, $\mu_{x_2} = 2$, variances $\sigma^2_{x_1} = 1$, $\sigma^2_{x_2} = 2$, and cov $(X_1, X_2) = 1$. Find

(a) the correlation coefficient of (X_1, X_2),

(b) $P\{X_1 - X_2 > 0\}$,

(c) $E(X_1^2 + X_2^2)$,

(d) $P\{\bar{X} > 1\}$.

3. Let X_1, X_2 be independent $N(\mu, 4)$. To test $H_0 : \mu = 0$ vs $H_1 : \mu > 0$ the UMP test of size $\alpha = .05$ is reject if $\bar{X} > C_\alpha$. Find C_α and find the power of the test when $\mu = 1$.

4. Let X_1, X_2 be a random sample of size 2 from a gamma distribution with parameters $\alpha = 3$ and θ. That is,

$$f_X(x; \theta) = \frac{e^{-x/\theta} x^2}{\Gamma(3)\theta^3}, \quad x > 0.$$

(a) Find a sufficient statistic T.

(b) Is $\log_e(X_1 + X_2)$ a sufficient statistic?

(c) Find the maximum likelihood estimator of e^θ.